

The provided audio explains the three-layer architecture of a Convolutional Neural Network (CNN). Since the Whiteboard OCR is empty, all content for these study notes is derived from the audio transcription.

Convolutional Neural Network (CNN) - Three-Layer Architecture

A Convolutional Neural Network (CNN) is primarily structured around a **three-layer architecture** designed to process and understand image data effectively. These layers work in sequence to extract features, reduce dimensionality, and make final predictions or classifications.

1. Convolutional Layer (???????????? ????)

This is the **first and foundational layer** of a CNN.

- * **Purpose:** To extract various features from the input image.
- * **Mechanism:**
 - * It uses a set of different **"filters"** (also known as kernels).
 - * Each filter is specifically designed to detect a particular feature within the image.
 - * These filters are slid across the image (convolution operation) to produce "feature maps."
- * **Example:**
 - * If you input an image of a person into the CNN:
 - * There would be a specific **"nose filter"** to detect the nose.

- * Another **"body filter"** to identify the body shape.
- * A **"hair filter"** to locate the hair.
- * By using various filters, the convolutional layer extracts a wide range of different features from the image.

2. Pooling Layer (?????? ????)

This layer typically follows one or more convolutional layers.

- * **Problem Addressed:** After performing convolutional operations, the resulting "feature maps" can be very large in size. This large size can lead to high computational costs and potential overfitting.
- * **Purpose:** To reduce the spatial dimensions (width and height) of the feature maps, thereby decreasing the computational load and making the model more robust to variations in the input.
- * **Types of Pooling:**
 - * **Maximum Pooling (???????? ????):** Selects the maximum value from a specific window (region) in the feature map, effectively capturing the most prominent feature.
 - * **Average Pooling (???? ????):** Calculates the average value from a specific window in the feature map.

The Feature Extraction Portion (???? ?????????????? ???? ??????)

The speaker highlights that the **sequential chain of Convolutional and Pooling layers working together** is referred to as the **"Feature Extraction Portion"** of the CNN. This is where the network learns to identify and distill meaningful patterns and characteristics from the raw input data.

3. Fully Connected Layer (?????? ????????)

This is the **final layer** in the CNN architecture, typically appearing after the feature extraction layers.

- * **Input:** It takes the highly processed and reduced feature maps (which are usually flattened into a 1D vector) from the preceding layers.

- * **Mechanism:**

- * In a fully connected layer, **every neuron is connected to every neuron in the previous layer.** This dense connectivity allows for complex relationships between the extracted features to be learned.

- * **Purpose:** To perform high-level reasoning, classification, or regression based on the features extracted by the convolutional and pooling layers. This layer makes the final decision or prediction of the network (e.g., classifying an image as "cat" or "dog").
